Certifying the Outcome of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Latency and Dataset Shift

Hassan Waqas Saleem, Rahat Zeeshan Saleem, and Hina Maryam

***Abstract*—A fallback trigger decides when an automated vehicle should abandon its task for a minimum-risk maneuver. Statistical certificates for such triggers bound the alert, not what happens once the maneuver executes. Because a rollout depends only on the tick at which the fallback fires, one reference run plus one forced rollout per decision tick evaluates every trigger, threshold and latency by lookup, checked against direct execution. On replayed Argoverse 2, nuScenes and Waymo logs with a rule-based driver and reactive traffic, we certify the alert with Learn-then-Test and the outcome, the harm in the run actually driven, with risk-ordered fixed-sequence testing. The alert certificate keeps its 5 % target (3.7 % missed, violations in 9.5 % of resplits against 43.5 % uncertified) but leaves 6.5 % harm after the fallback. Outcome certification at a 7.5 % target realizes 5.4 %. A certified time-to-collision trigger is then the cheapest, with 6 % unnecessary stops at 10 Hz and 13 % under 0.5 s of latency, so predictor-based triggers add nothing at these latencies. Only from 0.8 s does it fail to certify, and a learned or safe-distance trigger is needed. An apparent latency crossover at 2 Hz decisions did not survive a 10 Hz test. Certificates fail on nuScenes and in two of six held-out cities but hold on Waymo.**

***Index Terms*—Autonomous driving, conformal prediction, minimum-risk maneuver, distribution shift.**

I. Introduction

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk maneuver (MRM): slow to a stop, pull to the shoulder, or hand control back to a human. When to invoke it is, in most deployed and published systems, a hand-tuned threshold on a scalar confidence, uncertainty or risk score from a perception or prediction module. Such a threshold carries no guarantee, and learned uncertainty estimates are known to degrade under the shift a vehicle meets in a new city or weather. The failure is silent: the system keeps reporting a confidence number that is no longer right.

Conformal prediction offers a principled alternative, a distribution-free, finite-sample guarantee under exchangeability, and has been applied widely to trajectory forecasting and planning in driving. A guarantee on a prediction, or on a warning, is nevertheless not a guarantee on the consequences of the action it triggers. Whether an alert leads to a fallback is rarely evaluated in closed loop, and triggering is not free: a vehicle that stops in a live lane because its uncertainty grew is not obviously safer than one that continues, and a trigger that fires whenever asked trivially achieves a good miss rate while being useless. Reporting only coverage, or only collision rate, hides this trade-off.

We ask a narrower question than "which trigger is best": *can the outcome of a fallback trigger, not only its alert, be certified, and how does that certificate behave under actuation latency and dataset shift?* Certifying a threshold on a risk score against missed events is not new. Luo *et al.* [27] do so with conformal prediction, and Conformal Decision Theory [31] calibrates decisions through a long-run *average* guarantee. We build on these and do not claim the certification step. An MRM trigger nevertheless differs in kind: it is an irreversible, one-shot decision, its errors are paid in two currencies (missed interventions, and the harm of stopping itself), and a deployed vehicle needs its guarantee before the first mile. To our knowledge, no prior work certifies a fallback trigger and then measures, in closed loop, what the executed fallback does.

We organize the study around four research questions, each tied to a hypothesis fixed before data collection (Section IV-G). **RQ1 (H2):** does a certificate keep its promise on the closed-loop outcome, and what does it cost in unnecessary stops and induced collisions? **RQ2 (H1):** do open-loop metrics choose the trigger that closed-loop safety does? **RQ3:** how sensitive is it to the timing of the intervention, and can the outcome be certified under latency? **RQ4 (H3, H3-W):** what happens to a certificate under cross-dataset shift, and how much can unlabeled or a few labeled target scenarios repair? The outcome-level part of RQ3 was added by later amendments and is labeled as such.

We make four contributions.

1) **A lookup evaluation protocol for executed fallbacks.** Under a determinism assumption whose violation we measure, a rollout's outcome depends only on *when* the fallback fires (Observation 1, Fig. 1). One reference run plus one rollout forced to fire at each decision tick (about 17 per scenario) evaluates every trigger, threshold, online method and latency on identical physics, where direct execution would need over 7,000 rollouts per scenario here. We check the lookup against direct execution (Section III-E).

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2) **Outcome-level certification.** Risk-ordered fixed-sequence testing certifies the harm in the run the vehicle actually drives, a non-monotone loss that the alert-level certificate does not bound (it leaves 6.5 % harm after the fallback). At a 7.5 % target the certified threshold realizes 5.4 % harm with a valid violation frequency.
3) **A physics baseline that is hard to beat, and a latency map.** Under a time-to-collision (TTC) harm label, a certified TTC threshold is the cheapest trigger at the outcome level up to 0.5 s of latency with 10 Hz decisions (6–13 % unnecessary stops against 8–33 % for the safe-distance, geometric and learned triggers), and counterfactual labels are not needed for a learned trigger. Only from 0.8 s does TTC fail to certify. A latency crossover seen at 2 Hz decisions was an artifact, withdrawn after a pre-registered 10 Hz test.
4) **A failure map for the certificate.** An Argoverse 2 certificate fails on nuScenes and in two of six held-out cities but holds on Waymo, with failures tracking the target's harm rate. A noise-corrected violation measure separates methods that are clearly violated from those that are only near the boundary, and spatial dependence weakens the high-probability statement in practice.

Refuted hypotheses are reported as such, and analyses run after the pre-registered verdicts are labeled exploratory.

II. Related Work

A. Conformal prediction for autonomous driving

Conformal prediction [1], [9] gives prediction sets a distribution-free, finite-sample coverage guarantee under exchangeability and has been applied widely to trajectory forecasting and safe planning [10]: per scenario type [11], with feedback from realized trajectories [12], with CVaR clearance margins [13], and in crowd navigation [14]. Lindemann *et al.* [4] embed conformal regions in a model predictive controller, a guarantee that collapses under shift, which Rahaman *et al.* [23] address with robust conformal prediction; SafePath [24] guarantees that a set of LLM-planned paths contains a safe one. In these works the guarantee is on the *prediction* or a candidate set, not on a fallback-triggering decision replayed on driving logs under reactive traffic. Ours is on the decision, and in Section III-F on its executed outcome.

B. Online and decision-level conformal methods

Adaptive Conformal Inference (ACI) [32] updates the miscoverage level online, and Dixit *et al.* [5] use it inside a model predictive controller to size a prediction set, a guarantee on the region, not on a fallback decision. Conformal Decision Theory (CDT) [31] carries the idea to decisions: it updates $\lambda_{t+1} = \lambda_t + \eta(\varepsilon - \ell_t)$ so that the long-run average risk converges to ε, demonstrated on robot navigation with a continuous distance-based loss. Both give *asymptotic, average* guarantees. We target a one-shot, irreversible trigger whose certificate must hold with finite samples before deployment, and adopt both as baselines (Section IV-E). Offline risk control now covers non-monotone losses [15], several risks at once [16], and planning with control barrier functions [17]. Closest to our accounting of what the fallback does, Joshi *et al.* [18] certify the outcome of switching to a fallback policy, per instance and with no notion of timing. An MRM trigger is sequential and absorbing: *when* it fires determines every outcome after it.

C. Task-relevant and closed-loop failure detection

Farid *et al.* [25] propose a *task-relevant* monitor that flags a misprediction only when the resulting planning cost is anomalous, evaluated open-loop and semi-reactively without an actuated fallback. Their insight is reflected in our labeling: a collision suffered while stopped or struck from behind is an *induced-collision* cost of the fallback, never a missed intervention.

Closest to our certification step, Luo *et al.* [27] tune the threshold of a warning system with Mondrian conformal prediction so that, among situations that are unsafe in the true future *in the absence of corrective action*, fewer than ϵ raise no alert. Their predictor is Trajectron++ and their alerts are scored against logged data, so the alert triggers nothing. We adopt the same principle and do not claim it as new. The differences that matter are the guarantee (class-conditional there, marginal over scenarios here), the consequence of an alert (none there, an executed maneuver here), and dataset shift and latency, which they do not study (Supplementary Table S1). We implement their certificate as a baseline (Supplementary Section S1).

Sinha *et al.* [38] close the loop differently: a conformal runtime monitor detects when perception leaves the tolerance of a robust MPC controller that keeps a perception-free fallback plan feasible. Their fallback is part of the controller design, whereas ours is an irreversible maneuver whose trigger is the object of study. Atoui and Resende [19] design and validate an MRM controller, taking the decision to engage it as given. Scenario-based simulation is a standard route to evaluating automated vehicle safety [39], [40], needs a statistical foundation for stopping rules [20], and can use risk-based filtering [21]. We replay real logs with reactive traffic.

Fallback triggering also sits within established safety frameworks. Runtime monitors that switch to a trusted fallback, as in [19], [38], follow the runtime-assurance idea of a monitored learned component with a trusted backup. Switching policies have been learned for this purpose, with safety from reachability [45], but without a statistical certificate on the executed outcome, and conformal abort alerts have been calibrated in flight testing [46] without evaluating the response. Responsibility-Sensitive Safety [41] gives model-based rules for when a vehicle is at fault, whereas our trigger is certified from data. SOTIF [42] and UL 4600 [43] frame assurance at the system level, but do not specify how to certify a trigger threshold. Selective prediction [44] studies when a model should abstain at a controlled risk, the same reject-option structure as a trigger, except that here abstention is an executed maneuver with its own cost. We do not claim priority in any of these areas.

Chapman *et al.* [28] use Conditional Value-at-Risk to assess the severity of constraint violation, a tail extension we leave to future work. ScenarioNet [29] unifies real driving logs for

interactive replay in MetaDrive, which is how every scenario is replayed here.

III. Problem Formulation and Method

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ: at every decision tick the vehicle computes an uncertainty score $u_t(X)$ from a trajectory predictor's output for nearby agents, and engages the minimum-risk maneuver the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{1}\big[\text{harm in } X \text{ under } \lambda,\ \text{MRM not first}\big], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{1}\big[\text{MRM fires under } \lambda,\ X \text{ harm-free at } \lambda = \infty\big], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1,\ldots,X_n}\Big(\mathbb{E}_X\big[L(X, \hat{\lambda})\big] \le \alpha\Big) \ge 1 - \delta, \quad (3)$$

while minimizing $\mathbb{E}_X[S(X, \hat{\lambda})]$, the smallest autonomy loss compatible with the risk target α.

Let $h(X)$ be the first harm step of the reference run with the fallback disabled (∞ if none) and $\tau_\lambda(X)$ the firing step. Because a run under the trigger coincides with the reference run until it fires (Section III-E), $L(X, \lambda) = \mathbb{1}[h(X) \le \tau_\lambda(X)]$ and $S(X, \lambda) = \mathbb{1}[\tau_\lambda(X) < \infty,\ h(X) = \infty]$: both alert-level indicators depend only on the reference run and the firing step. The consequences of the fallback after it fires enter through separate quantities, which we report but do not certify: residual harm (harm anywhere in the executed run), induced collisions, and route completion. Residual harm is not monotone in λ, since firing earlier can itself create harm, and it includes harm after the alert, so (3) does not bound it. Section III-F certifies it directly. Since the running maximum of the score is non-decreasing, τ_λ is non-decreasing in λ and $L(X, \lambda)$ is non-decreasing in λ for every scenario. Equation (3) is therefore a monotone-risk statement, to which Conformal Risk Control [3] applies directly, but with an expectation guarantee. It bounds the calibration-averaged risk, not the probability that a given calibrated threshold exceeds α. We use Learn-then-Test [2] for the high-probability form. For a finite grid $\lambda_1 > \lambda_2 > \cdots > \lambda_J$ ordered from least to most conservative, we compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n}\sum_i L(X_i, \lambda_j)$ on n calibration scenarios and a Hoeffding–Bentkus p-value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, then apply fixed-sequence testing starting from the most conservative λ. Every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ, and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). The guarantee is therefore about the miss rate of the alert, over scenarios exchangeable with the calibration scenarios, with confidence $1 - \delta$ over the calibration draw. We report the violation frequency of both procedures over resplits rather than assuming it.

The marginal target is weaker than it may sound. With harm in a fraction π of scenarios, a marginal miss probability α leaves up to a fraction α/π of harmful scenarios unalerted, which is $32\,\%$ at $\alpha = 0.05$ and $\pi = 0.158$ in our data ($24\,\%$ observed, Section V-F).

B. Trigger score

At each decision tick the geometric score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle's own near-term extent. The ego's constant-heading, constant-speed extrapolation over T steps, $\{p_\tau^{\text{ego}}\}_{\tau=1}^{T}$, is the same route-agnostic motion model used by the harm metric (Section IV-B), deliberately not the planned path. Each predicted offset of agent i, mode k, is split into a longitudinal component $a_\tau^{(i,k)}$ and a lateral component $\ell_\tau^{(i,k)}$ with margins $m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion is

$$\rho_\tau^{(i,k)} = \max\left(0,\ 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}}\right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}}\right)^2}\right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. The length-aware box is a modeling choice made during development. Our ablation (Supplementary Section S14) finds no measurable benefit over a width-only isotropic radius, so we report it as a design option.

C. Fallback maneuver

On triggering, the vehicle executes the minimum-risk maneuver defined by UN Regulation No. 157 (Automated Lane Keeping Systems) § 5.5.1 [37]: decelerate inside the current lane with a deceleration demand not exceeding $4.0\,\mathrm{m/s^2}$, lane-keeping control otherwise unchanged, holding the vehicle stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0\,t,\ 0)$ from the moment of triggering.

D. Weighted repair under covariate shift (a heuristic)

The certificate in (3) assumes calibration and deployment scenarios are exchangeable. For covariate shift we reweight calibration scenarios by an estimated density ratio $w(x)$ from a domain classifier on pre-deployment covariates (agent counts and speeds, initial gap, route length) and unlabeled target scenarios, following weighted conformal prediction [33], and test the weighted risk with the Hoeffding–Bentkus p-value at the Kish effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. Because the weights are estimated and clipped and n_{eff} is an approximation, this is a heuristic repair with no finite-sample guarantee. As the weights concentrate, n_{eff} shrinks and the test certifies a more conservative threshold, and eventually none, which reports when the source data cannot support a guarantee for the target.

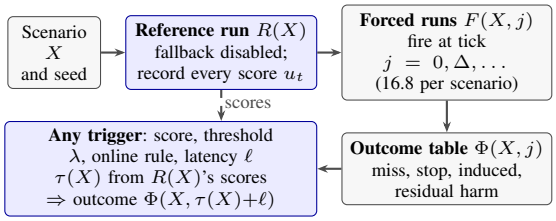


Fig. 1. Lookup evaluation (Observation 1). One reference run and one forced rollout per decision tick are simulated once per scenario. Every trigger, threshold, online update rule and latency is then evaluated by lookup on identical physics, instead of one rollout per combination (over 7,000 per scenario here).

E. Evaluating every trigger by lookup

Fix a scenario X and its random seed. The *reference run* $R(X)$ disables the fallback. For a decision tick j, the *forced run* $F(X, j)$ is identical to $R(X)$ until tick j and then executes the MRM for the rest of the episode. Let $\Phi(X, j)$ denote any outcome of $F(X, j)$ (miss, stop, induced collision, residual harm, route completion) and $\Phi(X, \infty)$ the outcome of $R(X)$. Fig. 1 summarizes the procedure.

Observation 1 (Firing-tick sufficiency): Assume (A1) the simulator, background traffic and predictor are deterministic functions of the seed and of the ego's trajectory, so two runs with the same seed and the same ego control up to a step have identical states up to that step; (A2) the trigger's decision at tick j depends only on the state history up to tick j; (A3) once the fallback fires it is absorbing, and the control after firing depends only on the state at firing. Then for every trigger with firing tick $\tau(X) \in \{0, \Delta, 2\Delta, \ldots\} \cup \{\infty\}$, the run under the trigger is $F(X, \tau(X))$, and every outcome equals $\Phi(X, \tau(X))$. This holds for any score, threshold, online update rule and actuation latency ℓ (for which the run is $F(X, \tau(X) + \ell)$).

Proof: Until τ the control is that of $R(X)$ by (A3) and the states equal those of $R(X)$ by (A1), so by (A2) the trigger's decisions, and hence τ, are determined by $R(X)$ alone. At τ the control depends only on the state at τ (A3), as in $F(X, \tau)$, and both runs have equal states afterwards by (A1). ∎

Two corollaries follow. Alert-level indicators need no forced run, since L and S are functions of $R(X)$ and τ alone, and the consequences of the fallback need only n_{ticks} forced rollouts per scenario (16.8 on average, at most 26), after which any number of triggers is evaluated by lookup, where direct execution needs over 7,000 rollouts per scenario here. The observation is elementary and its value is practical: an exhaustive, paired comparison becomes affordable. Assumption (A1) is an idealization, so we measure its violation. Over the first 33,761 forced rollouts (2,000 scenarios), 98.4 % agreed with the reference run to within 1 cm before firing, the pre-fire time-to-collision trace differed in 1.2 %, and the pre-fire harm status in one rollout (0.003 %). We report every analysis on all scenarios and after dropping the 3.6 % with any flagged drift. We also executed four triggers directly on 252 random test scenarios (1,008 scenario–trigger pairs, one with a one-tick latency). Firing step, miss, unnecessary stop and induced collision agreed with the lookup in all 1,008 pairs (upper bound 0.4 %), and harm anywhere in the run in all but one (Supplementary Section S7). The lookup is exact only under (A1), but it matched direct execution within this tolerance.

F. Outcome-level certification and learned triggers

The forced rollouts also give, for every scenario, the outcome of firing at each tick. Let the *safe window* $W(X)$ be the set of firing ticks (with "never") whose executed run has no harm, $F(X)$ its first tick, and $O(X, \lambda) = \mathbb{1}[\text{harm in the run under } \lambda]$ the *outcome loss*, with a total-harm variant that also counts induced collisions. O is not monotone in λ and $W(X)$ is empty when harm is unavoidable. Learn-then-Test needs a monotone order only to be powerful, not to be valid: any order fixed before looking at the calibration data yields a valid family-wise guarantee. We therefore order the threshold grid by the risk of O estimated out of fold on a training split, run fixed-sequence testing along that order on a separate calibration split, and take among the certified thresholds the one with the fewest unnecessary stops, which is valid because all certified hypotheses hold simultaneously. A threshold chosen on one split also yields a one-sided Clopper–Pearson upper bound on outcome harm from a fresh split.

We also learn triggers. A gradient-boosted classifier predicts, at each decision tick, from the stored scores, time-to-collision, headway distance, agent counts, predicted gaps and scenario covariates, whether firing is warranted, and its predicted probability is the trigger score. We compare two labels: a *counterfactual* label (the no-fallback run is harmful and $t \geq F(X)$, so firing is neither unnecessary nor too early) and a *factual* label (harm occurs within 2 s after the action takes effect). For a delay of ℓ the labels are shifted by ℓ, because a decision at t acts at $t + \ell$. All hypotheses on these triggers were pre-registered (Section IV-G).

IV. Experimental Setup

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [29] from Argoverse 2 Motion Forecasting [6] and nuScenes [7]. Non-ego traffic is controlled reactively by the Intelligent Driver Model [8], not replayed open-loop from the log. Pure log replay roughly tripled the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its log, precisely what a fallback maneuver produces, and would misrepresent the cost of stopping. The ego follows its logged route under IDM longitudinal control (MetaDrive's TrajectoryIDMPolicy) until the fallback engages, so the study concerns the trigger, not planning quality.

B. Harm labeling

A rollout is labeled harmful using the nuPlan closed-loop safety definition [30]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at 0.1 s resolution, not evaluated while the ego is stationary or against tracks behind it. Fault attribution follows the nuPlan convention. A collision the ego suffers while stationary, or

while struck from behind, is not at-fault and never counts as a missed intervention. It is recorded as an *induced collision*, the measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot [35], one of many learned multi-modal forecasters [22] (6 modes, 2.1 s history, 6 s horizon), trained with the UniTraj framework [36] on the Argoverse 2 Motion Forecasting training split (minADE_6 = 1.09 m on held-out validation scenes). It runs online inside the closed loop, so its input is the actual, possibly ego-deviated, simulator history, with the UniTraj preprocessing unchanged. Vehicles are predicted by the network. Pedestrians and cyclists, which the checkpoint was not trained on, receive constant-velocity forecasts.

D. Datasets and splits

All scenarios come from Argoverse 2 validation, disjoint from the predictor's training data, split deterministically by scenario identifier into development (14,990), calibration (4,971) and test (5,027) sets (design choices used the development split only). The shifted targets are nuScenes (9,041 scenarios of 8.1 s, Boston and Singapore) and, under a separate pre-registered amendment, 1,000 Waymo Open Motion [26] validation scenarios (9.1 s). Scenarios in which the ego moves less than 10 m are excluded, as in ScenarioNet. The pre-registered evaluation uses 2,000 calibration, 2,000 test and 1,500 nuScenes scenarios, of which ten Argoverse 2 scenarios with unmatched per-agent records are dropped from every analysis, leaving 1,994, 1,996 and 1,504. Decisions are taken at 2 Hz, with identical miss rates at 10 Hz on a paired development comparison.

E. Baselines

All methods use the same predictor, the same MRM and the same test scenarios, and are evaluated by lookup (Section III-E).

- **T0 — time-to-collision threshold** (physics trigger, no learned component): fire when the nuPlan time-to-collision, projected at constant speed and heading, falls below a threshold; certified with LTT or tuned. Headway distance and an RSS-style safe-distance trigger [41] (response time 0.5 s, accelerations 3, braking 4 and 8 m/s^2, fixed beforehand) are two more physics variants.
- **T1 — confidence threshold** (uncertainty practice): fire when $1-\max_k p_k$ of the network-predicted vehicle whose predicted modes come closest to the ego exceeds a threshold. Agents the network does not predict (pedestrians, cyclists, distant vehicles) are excluded from this score. The threshold is tuned on calibration data to the most permissive value with empirical miss rate $\leq \alpha$. A variant that uses the nearest agent ahead is also evaluated.
- **T2 — ensemble threshold**: the same tuning, applied to the spread of endpoint predictions across five Monte Carlo dropout passes [34].
- **T3 — open-loop conformal**: inflate every predicted mode by the split-conformal radius of the predictor's 3 s error on held-out Argoverse 2 logs at level α [1], and fire when the inflated region touches the ego. This is the guarantee-on-the-prediction design that the paper argues is insufficient.
- **Tuned geometric**: our score with an empirically tuned threshold and no certificate. It isolates the contribution of the calibrator.
- **CDT** [31] and **ACI** [32]: online decision- and prediction-level adaptation, updated after every scenario. Each uses the step size that performs best on test, which favors the baselines. Choosing it on calibration data instead gives CDT 4.6 % missed and 25.1 % stops, and ACI the same result.
- **Oracle**: the best threshold in hindsight on test data. It is not deployable and is reported as an upper bound.

F. Metrics and statistics

For every method we report the missed-intervention, unnecessary-stop and fallback-induced-collision rates, autonomy (the fraction of scenarios completed without the fallback) and route completion, with Wilson 95 % intervals. Validity of each certificate is assessed over 200 random half-and-half resplits, without replacement, of the pooled 3,990 calibration and test scenarios. A valid certificate keeps the frequency of test miss rate above α at or below δ. The resplits overlap, so they are not independent draws: the frequency estimates the violation probability for this pool, with Monte Carlo error (Wilson 95 % interval [6.2, 14.4] for 9.5 %, [36.8, 50.4] for 43.5 %). Scenarios from one place may be dependent. Resplitting whole spatial groups (ego starts within 150 m, a proxy) raises LTT's violation frequency to 17.0 % [12.4, 22.8], above δ, although group bootstrap intervals for the miss rates barely widen (Supplementary Section S12). Harm uses the nuPlan definition above, and Section V-H repeats the key comparison under three alternatives.

G. Pre-registration and reproducibility

The harm definition, $\alpha = 0.05$, $\delta = 0.10$, and the refutation criteria for three hypotheses were recorded in a version-controlled note before any calibration or test data were collected: **H1**, that open-loop metrics rank triggers differently from closed-loop outcomes; **H2**, that a certified trigger achieves a better safety–conservatism trade-off than a tuned fixed threshold; and **H3**, that a threshold certified on Argoverse 2 loses its guarantee on nuScenes. A later amendment, written before any Waymo outcome existed, added **H3-W**, the same hypothesis on Waymo, and a few-label recalibration study. The note carries further dated amendments (a corrected confidence baseline, the exactness measurements, robustness checks, a direct-execution check, and a strict-review amendment that added the TTC baseline, noise-corrected validity, outcome-level certification and the learned counterfactual trigger), each written before the corresponding outcomes were evaluated, and it labels the analyses run afterwards as post hoc. That last amendment was written after the main results

were known, and ten of its fourteen pre-registered claims were refuted (Supplementary Section S17).

Data and code availability. Code, the pre-registration note with all amendments, configuration files and per-scenario rollout records are available at https://github.com/hwaqas160/paper03-risk-fallback and archived at https://doi.org/10.5281/zenodo.23054382. Each scenario's record contains every score trace and outcome, so every table can be regenerated without re-simulation.

V. Results

All results use $\alpha = 0.05$ and $\delta = 0.10$, 1,994 calibration and 1,996 test scenarios from Argoverse 2, and 1,504 nuScenes scenarios, with Wilson 95 % intervals unless stated. Dropping the 3.6 % of scenarios with any flagged simulator drift changes no conclusion below. Sections V-A–V-C and the shift study are pre-registered, as is the Waymo recalibration (R1–R4). Section V-D and the TTC baseline come from a later amendment written after the main results were known, and the consequences of the intervention, the class-conditional comparison and the robustness checks are post hoc. Each is labeled.

A. Does a certificate keep its promise, and what does it cost? (RQ1)

Table I gives every method at its operating point on the Argoverse 2 test set. The certified geometric trigger (LTT) fails to pre-empt harm in 3.7 % of all scenarios [95 % CI 2.9–4.6], below the 5 % target, which is 24 % of the scenarios that are harmful, with 28.5 % unnecessary stops and 2.6 % fallback-induced collisions. The last column repeats the calibration on 200 random resplits. A valid certificate keeps the uncorrected frequency at or below $\delta = 10\,\%$, and LTT does (9.5 %). That count also includes test-sample noise, so we add a corrected count that only counts resplits whose test miss is clearly above α. Empirical tuning, Conformal Risk Control (CRC), which guarantees only an expectation, and the tuned confidence trigger are clearly violated under both counts (43.5 %, 41.0 % and 50.5 % uncorrected; 12–15 % corrected), while LTT is not violated under either. Among the predictor-based triggers, certification is therefore the only design that reliably keeps its target at a useful operating point (the uncertified TTC threshold sits far below the target and is only indeterminate). Fig. 2 plots every method against the geometric score's own curve as its threshold varies, and Fig. 3 shows the resplit distributions behind the violation column. CRC and empirical tuning select the same threshold here, because no grid point falls between the calibration risks 4.81 % and 5.02 %.

The certificate has a price, and it does not win the trade-off. The tuned geometric threshold and CDT both stop less often than LTT (paired bootstrap difference -2.5 points, 95 % CI $[-3.2, -1.8]$ against tuned). Hypothesis H2 required no baseline to be within 3 points of the certified trigger. It is **refuted**. The supportable claim is narrower: among the designs that keep their promise at a useful operating point, the certified one is the only one that reliably meets it. Open-loop conformal never violates, but fires in 73.8 % of scenarios.

TABLE I
Closed-loop outcomes on the Argoverse 2 test set at $\alpha = 0.05$. Rates in %, unnecessary-stop rate with its 95 % Wilson CI where available. Last column: fraction of 200 calibration/test resplits in which the test miss rate exceeds α (u, uncorrected), and in which its one-sided 95 % lower confidence bound does (c, corrected). Valid if u $\leq$ 10. CDT and ACI: 100 resplits.

Method	Miss	Unnec.	95 % CI	Induced	Auton.	Viol. u/c
No fallback	15.1	0.0	[0.0, 0.2]	0.0	100	–
Always fire	0.0	84.9	[83.2, 86.4]	6.5	0	–
T0 TTC, LTT (certified)	3.1	7.1	[6.1, 8.3]	1.0	78.3	1.0 / 1.0
T0 TTC, tuned	3.5	5.7	[4.7, 6.8]	0.9	79.8	50.5 / 1.0
T1 confidence	4.6	68.3	[66.2, 70.3]	4.0	20.3	50.5 / 14.0
T2 MC-dropout spread	4.0	70.8	[68.8, 72.7]	4.9	17.6	–
T3 open-loop conformal	0.0	73.8	[71.8, 75.7]	5.5	11.1	0.0 / 0.0
Tuned geometric (no cert.)	4.3	26.0	[24.1, 28.0]	2.3	61.4	43.5 / 15.0
CRC (geometric)	4.3	26.0	[24.1, 28.0]	2.3	61.4	41.0 / 12.0
LTT (geometric, certified)	3.7	28.5	[26.5, 30.5]	2.6	58.5	**9.5 / 0.0**
CDT (online)	4.9	24.1	–	2.2	63.9	59.0 / 0.0
ACI (online)	1.3	45.0	–	3.5	40.2	0.0 / 0.0
Oracle (hindsight)	5.0	23.6	–	2.2	64.2	–

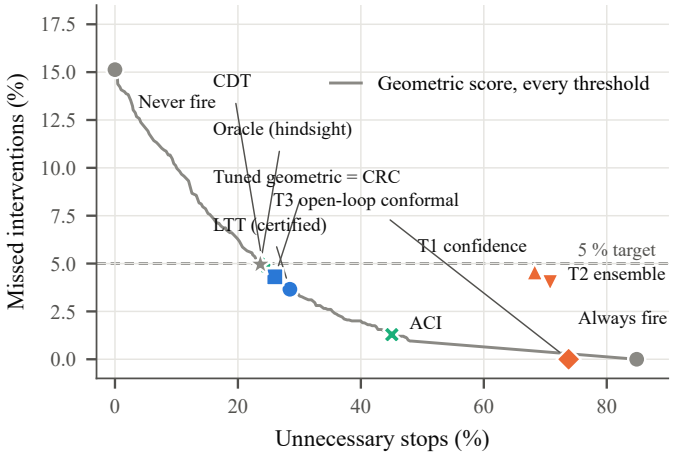


Fig. 2. Every method at its operating point (Table I), against the geometric score's own miss-versus-stops curve as its threshold varies. On this split most methods sit below the dashed $\alpha = 0.05$ line; the geometric methods do so at well under half the stops of the confidence, ensemble and open-loop triggers. Which of them keeps the target across resplits is shown in Fig. 3.

A physics trigger dominates at zero latency. A time-to-collision threshold, certified with the same procedure, meets the target with 3.1 % missed and only 7.1 % unnecessary stops and 1.0 % induced collisions (Table I), against 28.5 % for the predictor-based geometric trigger. The comparison is not neutral, because the harm label is itself defined by time-to-collision, and it holds in three of four alternative harm definitions, including collision only (Supplementary Section S14). Under this label the predictor adds nothing at zero latency, and the confidence and ensemble triggers, which stop 40–45 points more often than the geometric trigger, are weak comparators, in part because this predictor's mode probabilities are weakly informative (AUROC 0.63) and miscalibrated (ECE 0.25). ACI (1.3 % at 45 %) and open-loop conformal (0.0 % at 73.8 %) show a guarantee on the prediction not translating into a useful decision.

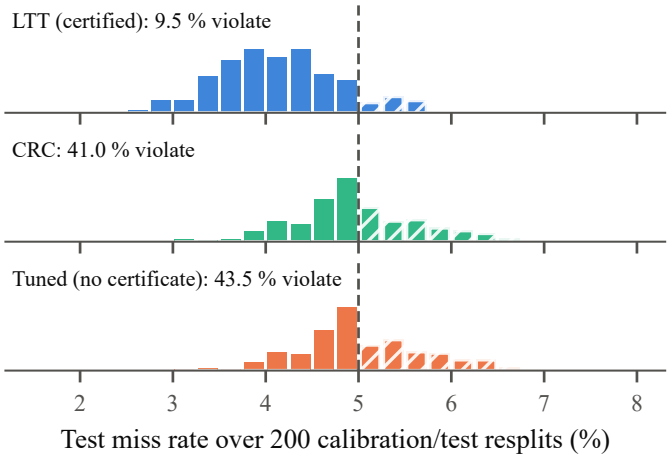


Fig. 3. Test miss rate over 200 calibration/test resplits. Dashed line: the 5 % target; hatched bars: resplits that violate it. A valid certificate keeps violations at or below $\delta = 10\,\%$. LTT's high-probability guarantee does (9.5 %); CRC's expectation guarantee and empirical tuning do not.

B. Do open-loop metrics choose the closed-loop trigger? (RQ2)

Ranking six trigger scores by open-loop AUROC of their scenario peak and by closed-loop unnecessary stops at each certified threshold (Supplementary Section S14), the rankings agree at the top and diverge below it: Kendall's $\tau = 0.60$ on Argoverse 2, 0.47 on nuScenes and 0.73 on Waymo. The pre-registered rule refutes H1 only if $\tau \geq 0.8$ and the top score coincides in every setting, so H1 is not refuted, but the evidence is weak. The clearest disagreement is at the bottom: the confidence score has AUROC 0.45, below chance, yet a threshold on it still reaches the miss target at 75.8 % stops, by firing indiscriminately.

C. Ablations and latency (RQ3)

Ablating the geometric score (Supplementary Section S14), a route-following corridor makes the trigger fire in essentially every scenario (84.9 % stops), consistent with the route reaching conflict points that the kinematic harm label does not count, replacing geometry with confidence or ensemble spread costs 44–47 points of stops, and a width-only isotropic radius performs the same as the length-aware box (27.5 % versus 28.5 %). With decisions at 2 Hz, one tick (0.5 s) of delay lifts the alert-level miss rate above the target (5.3 %) and two ticks give 9.1 % (Fig. 4b), and the zero-latency certificate is clearly violated at both (53 % and 100 % corrected). Calibrating the alert under the deployed latency gives 3.6–4.1 % missed at 34.4 % and 42.5 % stops and is not violated under the corrected count (1.5 % and 0.5 %; its uncorrected counts, 12.0 % and 11.5 %, are ambiguous).

D. Outcome-level certification, with and without latency

The alert certificate leaves 6.5 % harm after the fallback. We therefore certify the residual harm of the executed run at 7.5 % (no fallback leaves 15.1 %, always firing 4.1 %, so 5 % is barely reachable) with risk-ordered fixed-sequence testing

TABLE II
UNNECESSARY STOPS (%) OF THE OUTCOME-CERTIFIED THRESHOLD AT A 7.5 % RESIDUAL-HARM TARGET ON THE 10 HZ ARM (CALIBRATE 441, TEST 884; 2.2–2.5 % OF SCENARIOS THAT DID NOT REPRODUCE THE STORED REFERENCE RUN ARE DROPPED). REALIZED RESIDUAL HARM IS AT MOST 6.8 % IN EVERY CELL. †: ALWAYS FIRING. GEOMETRIC: PREDICTOR SCORES HELD AT 2 HZ. LEARNED (CF.), COUNTERFACTUAL LABEL; LEARNED (FACT.), FACTUAL LABEL.

Latency	TTC	Safe dist.	Geometric	Learned (cf.)	Learned (fact.)
0 s	**5.8**	20.1	32.9	19.0	7.6
0.2 s	**8.7**	24.7	28.5	25.2	17.6
0.3 s	**9.5**	23.9	26.5	28.4	20.4
0.5 s	**12.6**	23.6	31.9	25.6	30.7
0.8 s	86.7†	**29.9**	86.7†	36.5	31.4
1.0 s	86.7†	40.2	86.7†	**30.9**	39.6

(Section III-F). At zero latency on the 2 Hz arm (calibrate 904, test 2,900) every score certifies and meets the target. TTC (10.2 % unnecessary stops) and a supervised time-to-harm trigger (9.6 %) are the cheapest, ahead of the counterfactual-label trigger (13.6 %) and the geometric score (32.8 %), so the pre-registered claims that the counterfactual-label trigger beats TTC, and beats a supervised time-to-harm trigger, are both **refuted**.

Latency. On the 2 Hz arm a one-tick delay is 0.5 s with no chance to re-decide, and TTC then needed 85 % unnecessary stops against 26 % for the learned trigger. This crossover did not survive the test we pre-registered against it, a new arm with a forced rollout at every simulator step (10 Hz decisions; Table II). There TTC stays the cheapest through 0.5 s, and the learned counterfactual-label trigger stops 13–19 points more (paired CIs exclude 0) and is no better than the safe-distance trigger (**refuted** at 0.3 and 0.5 s). The 2 Hz result came from a certification cliff: the 7.5 % target lies at the risk of a 15 %-stop threshold, so the certified set flips to always firing in some calibration samples (Supplementary Section S20). Only at 0.8 and 1.0 s (post hoc) does TTC fail to certify (87 %), while the safe-distance and learned triggers need 30–40 %; at 1.0 s the learned trigger is cheapest (9.3 points fewer than the safe-distance trigger, CI [5.7, 12.9]). Physics features alone give the same learned trigger as all features (24.8 % against 25.5 % at 0.5 s on the 2 Hz arm), so the predictor adds no lead time beyond physics here.

E. Dataset shift (RQ4)

The Argoverse 2 certificate does not transfer. On nuScenes the threshold certified at a 5 % target misses 13.2 % [11.6, 15.0], 2.6 times the target and with an interval that excludes it. The pre-registered hypothesis H3 is **supported** (15.2 % after dropping drift scenarios). Part of this is a floor rather than shift: 5.0 % of nuScenes scenarios already contain harm at the first decision tick, which even a trigger that always fires cannot prevent (0.1 % on Argoverse 2). That floor equals the target, so no threshold can be certified at $\alpha = 0.05$ on nuScenes. With those scenarios excluded (post hoc), the shifted miss rate is 9.2 % [7.8, 10.8], still above the target. These intervals treat scenarios as independent, but the 1,504 nuScenes scenarios come from only 37 scenes. Resampling scenes widens the interval to [6.1, 21.0], which still excludes the target, while a

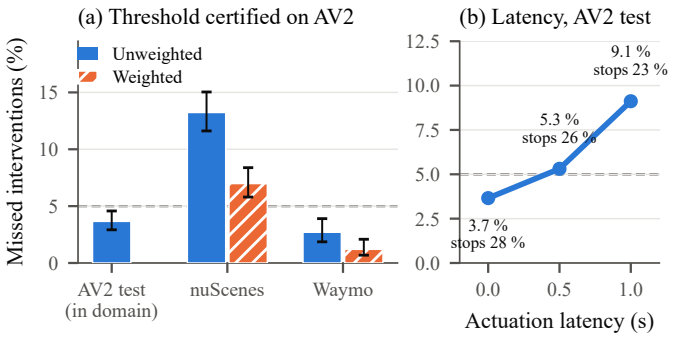


Fig. 4. (a) Miss rate of the Argoverse 2-certified threshold deployed unchanged on nuScenes and Waymo, unweighted and with the density-ratio repair (Section III-D); error bars are 95 % Wilson CIs. (b) Miss rate and unnecessary stops as actuation latency increases (Supplementary Section S14).

TABLE III
A THRESHOLD CERTIFIED ON ARGOVERSE 2 (LTT, $\alpha = 0.05$) DEPLOYED UNCHANGED ON NUSCENES AND WAYMO. "AVOIDABLE" EXCLUDES THE 5.0 % OF NUSCENES SCENARIOS WITH HARM AT THE FIRST DECISION TICK (POST HOC). WEIGHTED: DENSITY-RATIO REPAIR, NO TARGET LABELS.

Setting	Miss (%) [95 % CI]	Unnec. (%)	n
Argoverse 2 test (in domain)	3.7 [2.9, 4.6]	28.5	1,996
nuScenes, unweighted	13.2 [11.6, 15.0]	26.1	1,504
nuScenes, weighted	7.0 [5.8, 8.4]	52.4	1,504
nuScenes, oracle threshold	5.0	77.9	1,504
Avoidable, unweighted	9.2 [7.8, 10.8]	27.1	1,429
Avoidable, weighted ($n_{\text{eff}} = 68$)	2.2 [1.6, 3.1]	54.5	1,429
Waymo, unweighted	2.7 [1.9, 3.9]	45.1	1,000
Waymo, weighted	1.2 [0.7, 2.1]	50.2	1,000
Waymo, oracle threshold	5.0	33.9	1,000

second, disjoint set of 23 scenes gives 9.5 % with scene-level interval [3.9, 16.5], which does not: the direction of the failure replicates but its size is uncertain (Supplementary Section S6). A domain classifier on the covariates separates Argoverse 2 from nuScenes (cross-validated AUROC 0.97) far more easily than from Waymo (0.91), so the certificate failed under the larger measured shift and held under the smaller. Six further natural shifts, the Argoverse 2 cities held out one at a time, add evidence: the certificate met the target in four of six (67 %, below the 80 % we pre-registered, so that claim is refuted) and failed in Miami (6.1 % [4.8, 7.8]) and Washington, DC (5.3 %), the two cities with the highest harm rates. Across these and the two datasets the held-out miss rate tracks the target's harm rate (Spearman 0.90, $n = 8$, post hoc), but estimating that rate from unlabeled alerts to correct the target was unreliable and did not help (Supplementary Sections S10 and S15).

Density-ratio weighting moves the miss rate from 13.2 % to 7.0 %, still above the target, and doubles unnecessary stops (26.1 % to 52.4 %). On the avoidable subset it reaches 2.2 %, but with an effective sample size of only 68, almost entirely through conservatism (54.5 % stops).

F. What the intervention does, and a class-conditional certificate (post hoc)

The certified quantity is the alert. At the certified threshold on Argoverse 2, 3.7 % of scenarios are missed, in a further 1.7 % the fallback fires before harm but harm still occurs, in 9.8 % harm is prevented, in 1.1 % of the harmless scenarios the intervened run contains harm, and 2.6 % suffer an induced collision (Supplementary Section S19). Harm in the intervened run is therefore 6.5 % (15.1 % without a fallback), almost twice the certified miss rate; on the fresh split it is 5.0 % [3.7, 6.6] against 3.4 % missed. A trigger that always fires creates harm in 3.1 % of scenarios, and on nuScenes the same threshold leaves 15.6 %. Induced collisions are other vehicles striking the ego, at low to moderate speed (median speed difference 2.4 m/s), typically 6.5 s after firing and often where it starts at standstill, so this cost may partly reflect simulated traffic that does not yield to a stationary vehicle (Supplementary Section S6).

Class-conditional certification. A Luo-style class-conditional certificate is equivalent to always firing at $\epsilon = 0.05$ and misses 66 % of harmful nuScenes scenarios at a 32 % target (Supplementary Section S1).

G. Second shifted dataset and recalibration

Waymo Open Motion. Table III and Fig. 4a deploy the same Argoverse 2-certified threshold, unchanged, on 1,000 Waymo Open Motion validation scenarios [26] (9.1 s, 1 s of history). It misses 2.7 % [1.9, 3.9], below the 5 % target. Hypothesis H3-W was pre-registered with the expectation that this interval would exclude α from above, as on nuScenes; it does exclude α, but from below, so H3-W is **refuted** and failure under shift is dataset-dependent. A density-ratio-weighted repair, applied for consistency even though nothing here needed repairing, lowers the miss rate further to 1.2 % [0.7, 2.1] at a small increase in stops (45.1 % to 50.2 %).

Waymo recalibration (pre-registered R1–R4). Drawing k labeled Waymo scenes at random (200 draws; Supplementary Section S3), uncertified tuning violates the target in 69.5 % and 59 % of draws with 40 and 60 target scenes (**R1 holds**). Weighting the Argoverse 2 calibration set with 60 Waymo scenes ($n_{\text{eff}} = 232$) keeps validity (0 % violation, **R2 holds**) at a cost of 21.3 points of unnecessary stops (**R3 holds**). **R4**, which restates H3-W, is refuted: the source certificate already holds on Waymo. Few labeled target scenes are dangerous without a certificate even when the source certificate transfers.

Sensitivity. Non-reactive replay traffic, a 2.0 m/s^2 comfort deceleration and 10 Hz decisions leave the certified miss rate below the target (Supplementary Section S4). Thresholds certified on reactive traffic and deployed on replay traffic miss 3.7 % [2.2, 6.1] with 35.4 % stops against 71.7 % for the tuned confidence trigger (Section S11).

H. Robustness checks (post hoc)

Table IV collects the checks that vary the data, predictor, traffic, harm definition or splitting, each on data not used above and pre-registered as post hoc before running. The validity of Learn-then-Test and the advantage of the geometric score over the confidence and ensemble triggers survive all of them. The TTC trigger beats the geometric score under three of the four harm definitions, so the geometric advantage is over uncertainty-based triggers only. The cities and spatial groups are where the guarantee weakens (Section V-E).

TABLE IV
ROBUSTNESS AND REPLICATION CHECKS. VIOL.: RESPLIT VIOLATION FREQUENCY OF LTT, UNCORRECTED / CORRECTED; UNNEC.: UNNECESSARY STOPS OF LTT AGAINST THE TUNED CONFIDENCE TRIGGER (T1). RATES IN %.

Check	Setting	Result
Lookup vs direct execution	252 scenarios, 4 triggers	0 of 1,008 alert-level disagreements
Fresh disjoint data	904 / 904 scenarios	Viol. 7.5 / 0.5, miss 3.4
Second predictor	AutoBot, minADE 0.854	Viol. 9.0 / 0.0, stops 30.6 vs 68.3
Second architecture	Wayformer	Viol. 7.0 / 0.0, stops 41.8 vs 67.8
Non-reactive traffic	log replay	Miss 3.7 [2.2, 6.1], stops 35.4 vs 71.7
Decision rate	10 Hz, 0.2–0.5 s	TTC stays cheapest (9–13 stops)
Alternative harm	collision only, TTC 0.5 s, 1.5 s	Geometric beats T1/T2 by ≥ 20 points
Held-out cities	6 Argoverse 2 cities	Target met in 4 of 6
Spatial groups	150 m	Viol. 17.0 / 1.0, indeterminate

VI. SCOPE AND FUTURE WORK

What we claim, and what we do not. We claim that certification makes a closed-loop trigger keep its stated miss rate, that the outcome of the executed fallback can be certified with risk-ordered testing, that a certified physics trigger is the one to beat, and that the guarantee is fragile to shift and dependence. We do not claim that a learned or predictor-based trigger is better at the latencies studied, a guarantee under shift, or one on outcomes other than those calibrated.

The harm label and the trigger. Harm is defined by time-to-collision, so a TTC trigger is close to the label and its advantage is partly structural. Collision-only harm weakens it at the alert level, and outcome-level certification under collision-only harm was underpowered (Supplementary Section S20). Predictor-based triggers matter only when latency is long enough that TTC cannot certify, and a learned trigger needs no counterfactual labels. The supervised vehicle is a rule-based driver, not a learned planner, so results concern triggering.

Alert versus outcome. The alert certificate leaves 6.5 % harm after the fallback and 24 % of harmful scenarios unalerted (58 % on nuScenes). The outcome certificate of Section III-F bounds the executed run's harm, but only as well as the calibration data resemble deployment. The 5.0 % first-tick floor on nuScenes shows that a miss metric on harm anywhere in a replayed snippet mixes preventable events with events already under way, and we propose an avoidable-only variant for future benchmarks.

Simulator, predictor, perception and dependence. All results use MetaDrive with rule-based (IDM) agents, so whether the ordering of triggers survives learned agent behavior is open, and the induced-collision rate is best read as an upper bound. The geometric trigger's validity and its gap to the uncertainty triggers replicate with a second AutoBot and with Wayformer (Supplementary Section S13). The predictor receives agent states, which isolates triggering from perception error. All guarantees treat scenarios as exchangeable, which nearby scenarios may violate: our nuScenes samples span only 37 and 23 scenes, and spatial-group resplits leave the high-probability statement indeterminate.

VII. CONCLUSION

A certificate on a fallback trigger bounds what it was calibrated to bound, and what that is matters. With the fallback executed, a Learn-then-Test certified alert keeps its 5 % target in distribution (3.7 % missed, violations in 9.5 % of resplits against 43.5 % uncertified) but leaves 6.5 % harm after the fallback, and risk-ordered testing certifies the executed outcome directly (5.4 % realized against a 7.5 % target). Under a time-to-collision harm label a certified time-to-collision threshold is the cheapest trigger up to 0.5 s of latency with decisions at 10 Hz (6–13 % unnecessary stops), so predictor-based and learned triggers add nothing there and pay off only at 0.8 s or more, where TTC fails to certify. A latency crossover seen at 2 Hz decisions was an artifact and was withdrawn. The certificate fails on nuScenes and in two of six held-out Argoverse 2 cities, both with high harm rates, but holds on Waymo. For the setting studied, these findings argue for certifying the outcome at the deployed latency and decision rate, auditing it on data that resemble deployment, and checking a physics baseline first; the lookup protocol makes each inexpensive to repeat.

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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